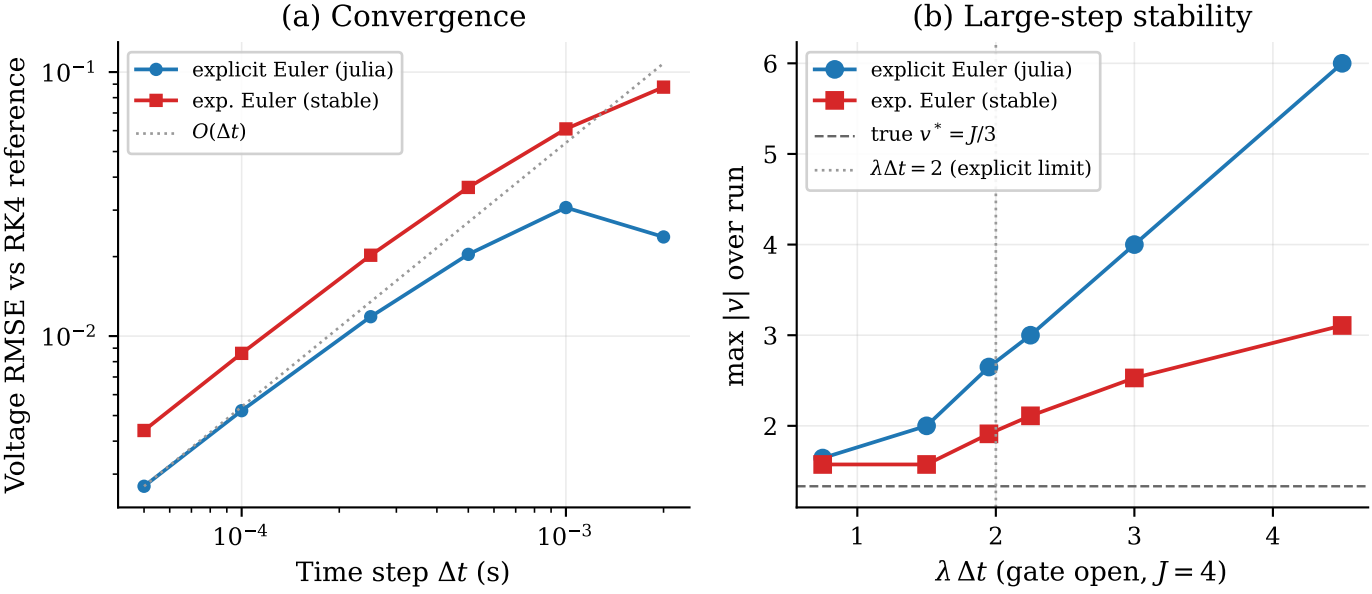
(a) Convergence
explicit Euler (julia)
exp. Euler (stable)
$O(\Delta t)$
Voltage RMSE vs RK4 reference
10^{-1}
10^{-2}
10^{-4}
10^{-3}
Time step Δt (s)
(b) Large-step stability
explicit Euler (julia)
exp. Euler (stable)
true $v^* = J/3$
$\lambda\Delta t = 2$ (explicit limit)
max $|v|$ over run
2
3
4
5
6
1
2
3
4
$\lambda\,\Delta t$ (gate open, $J = 4$)